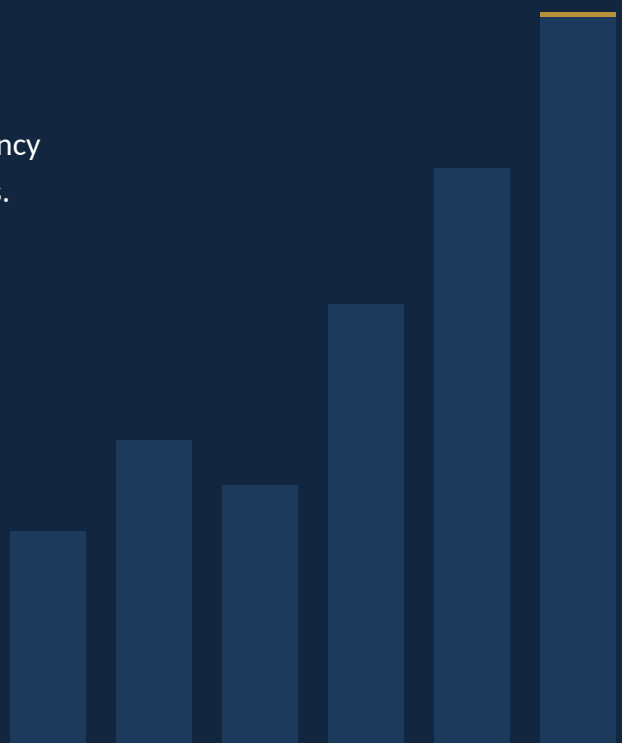
RETAIL SALES ANALYSIS

Business Insights Report

Reporting period	**2022–2025**
Dataset size	**18,045 records · 36 columns**
Analysis environment	**Microsoft SQL Server**

PURPOSE

Evaluate sales performance, profitability, customer behaviour, product contribution and operational efficiency — then translate findings into practical business actions.



Prepared as a SQL-based business analysis portfolio report.

1. Executive Summary

This project analyses retail sales data from 2022 to 2025 to evaluate business performance, profitability, customer behaviour, product contribution, sales channels, discounts, returns and operational efficiency.

The analysis identifies key performance patterns, highlights areas for improvement and translates findings into recommendations for sustainable growth.

Major findings

- Sales declined in 2024 compared with 2023, followed by a strong recovery in 2025.
- Electronics is the largest revenue-generating product category.
- Online and Retail Store channels are major contributors to sales.
- Higher discount bands are associated with lower average profit per transaction.
- Returns, delivery performance and inventory management present opportunities for operational improvement.

> **Interpretation note:** the report describes patterns in the available dataset. It does not establish that a particular factor caused a change in sales.

2. Dataset Overview

The dataset contains retail transaction records covering 2022–2025.

Attribute	Details
Dataset	Retail Sales
Time Period	2022–2025
Total Records	18,045
Total Columns	36
Database	Microsoft SQL Server
Analysis Tool	SQL

Main columns used in the analysis

Transaction & Customer:	Transaction ID, Order ID, Customer ID, Customer Segment
Date & Time:	Order Date, Order Time, Order Year
Sales & Profit:	Sales Amount, Cost Amount, Profit, Quantity, Unit Price, Discount Percentage
Product:	Product ID, Product Name, Product Category, Product Subcategory
Geography:	Country, Region, City, Store Name
Operations:	Order Status, Shipping Method, Delivery Days, Return Flag, Return Reason
Other Columns:	Payment Method, Sales Channel, Customer Rating, Inventory Level, Promotion Code, Sales Representative, and other customer and store details.

3. Key Business Metrics

The following KPIs provide an overview of overall business performance.

TOTAL SALES	TOTAL PROFIT	TOTAL COST	PROFIT MARGIN	UNIQUE CUSTOMERS
$8.02M	$1.64M	$6.42M	~20%	4,882

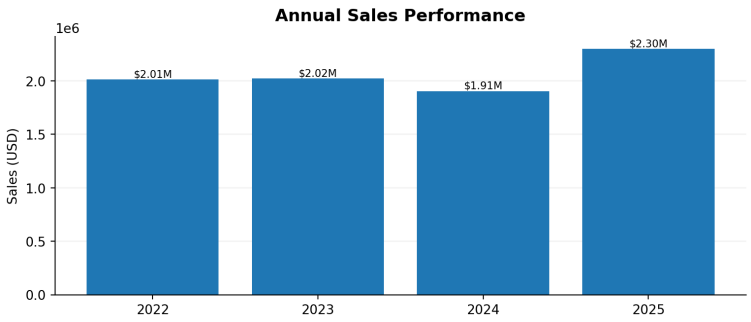
UNITS SOLD	AVG. SALES / TRANSACTION	AVERAGE RATING	AVG. DELIVERY TIME	TRANSACTION RETURN RATE
38,277	$445.60	3.99 / 5	2.26 days	10.40%

Currency is displayed as USD ($) based on the assumed dataset convention; the currency is not explicitly confirmed by the supplied column names.

4. Sales & Profitability Performance

Annual sales remained relatively stable between 2022 and 2023, declined in 2024 and increased significantly in 2025.

Year	Total Sales
2022	$2.01M
2023	$2.02M
2024	$1.91M
2025	$2.30M



Key insights

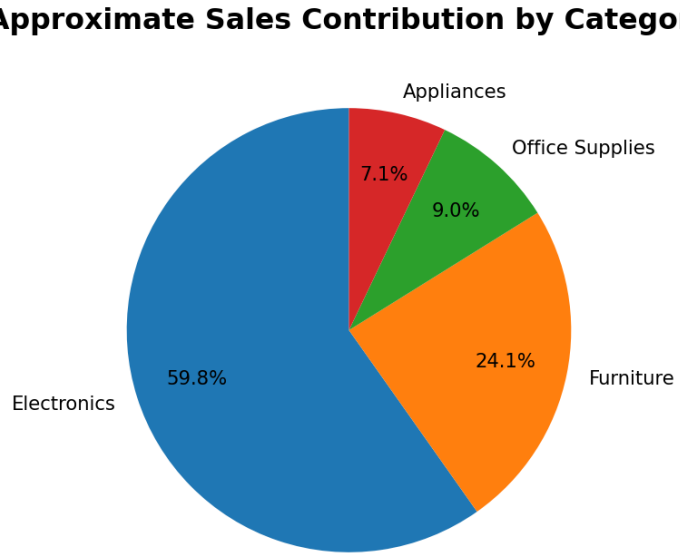
- Sales were almost unchanged between 2022 and 2023.
- 2024 experienced a decline compared with 2023.
- 2025 recorded the highest annual sales in the dataset.
- The 2025 result indicates a strong revenue recovery.

Business recommendation: Investigate the factors behind the 2024 decline and identify the products, regions and channels associated with the 2025 recovery.

5. Product & Customer Analysis

Product Category Contribution

Electronics is the largest revenue-generating category, followed by Furniture.



- Electronics contributes approximately 59.8% of total sales.
- Furniture contributes approximately 24.1%.
- Office Supplies and Appliances contribute smaller shares.
- Laptops are among the strongest-performing product subcategories.

Product recommendations: Maintain sufficient inventory of high-performing electronics; explore opportunities to increase Furniture sales; review weaker products for pricing and demand issues; and promote complementary products to increase customer spending.

Customer Analysis

The Consumer segment contributes the highest revenue, followed by Small Business and Returning Customer segments.

- Consumer customers contribute approximately 44.51% of sales.
- Small Business contributes approximately 17.26%.
- Returning Customer contributes approximately 13.54%.
- Premium customers contribute a smaller share of revenue.

Customer recommendation: Focus on retention, targeted offers and loyalty programmes. Analyse purchasing patterns to identify opportunities for repeat purchases and higher customer lifetime value.

6. Geographical & Sales Channel Analysis

Geographical Performance

The United States is the largest contributor to total sales.

Country	Sales
United States	$3.51M
United Kingdom	$1.33M
Canada	$1.20M
Germany	$965.97K
France	$727.62K
Australia	$513.30K

- The United States contributes approximately 42.54% of revenue.
- The UK and Canada are other important markets.
- Australia has the lowest sales contribution.
- Germany experienced a notable decline in 2024.

Recommendation: Strengthen operations in high-performing markets while investigating weaker markets to identify opportunities for regional expansion.

Sales Channel Performance

Channel	Sales
Online	$3.12M
Retail Store	$2.45M
B2B Portal	$2.13M
Phone Order	$552.91K

- Online is the largest sales channel.
- Retail Stores and B2B Portal also contribute significantly.
- Phone Orders contribute the smallest share.

Recommendation: Invest in online customer experience while improving physical retail store and B2B performance.

7. Discount, Returns & Operations

Discount Analysis

The analysis indicates an association between higher discounts and lower average profit per transaction.

Discount Range	Average Profit
0%	$120.69
0–5%	$97.27
5–10%	$74.88
10–15%	$50.44
15–20%	$24.11
20–25%	$1.83
Above 25%	-$6.83

- Average profit decreases as discount levels increase.
- Discounts above 20% leave very little average profit.
- Discounts above 25% are associated with negative average profit.

Recommendation: Introduce discount limits based on product margins. Evaluate promotions using incremental profit rather than sales revenue alone.

Return Analysis

Return Metric	Records
Returned transactions	1,877
Non-returned transactions	16,168
Transaction-level return rate	10.40%

Major Return Reasons

Return Reason	Records
Late Delivery	331
Wrong Item	320
Other	312
Damaged	312
Defective	302
Changed Mind	300

Recommendation: Improve delivery reliability, product quality checks and order fulfilment accuracy. Track return reasons regularly to identify recurring operational problems.

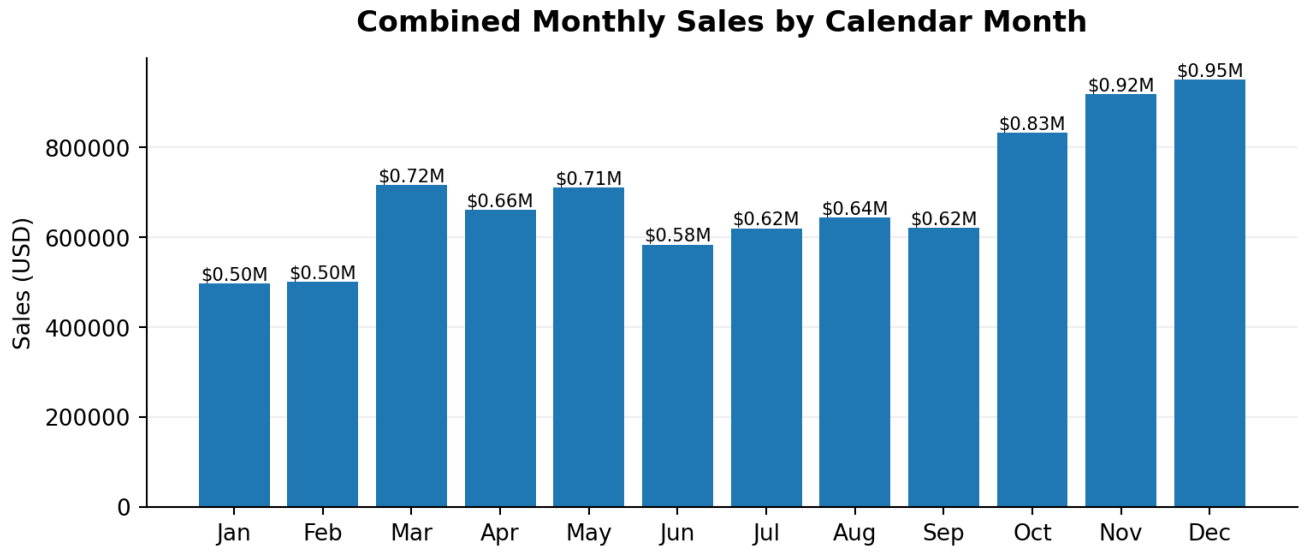
Shipping Performance

Shipping Method	Average Delivery Days
Pickup	0
Next Day	1.37
Express	1.85
Standard	4.53

Recommendation: Review Standard shipping performance and identify opportunities to reduce delivery delays without significantly increasing shipping costs.

8. Seasonality & Demand Analysis

Monthly sales vary throughout the year, with stronger performance towards the final quarter.



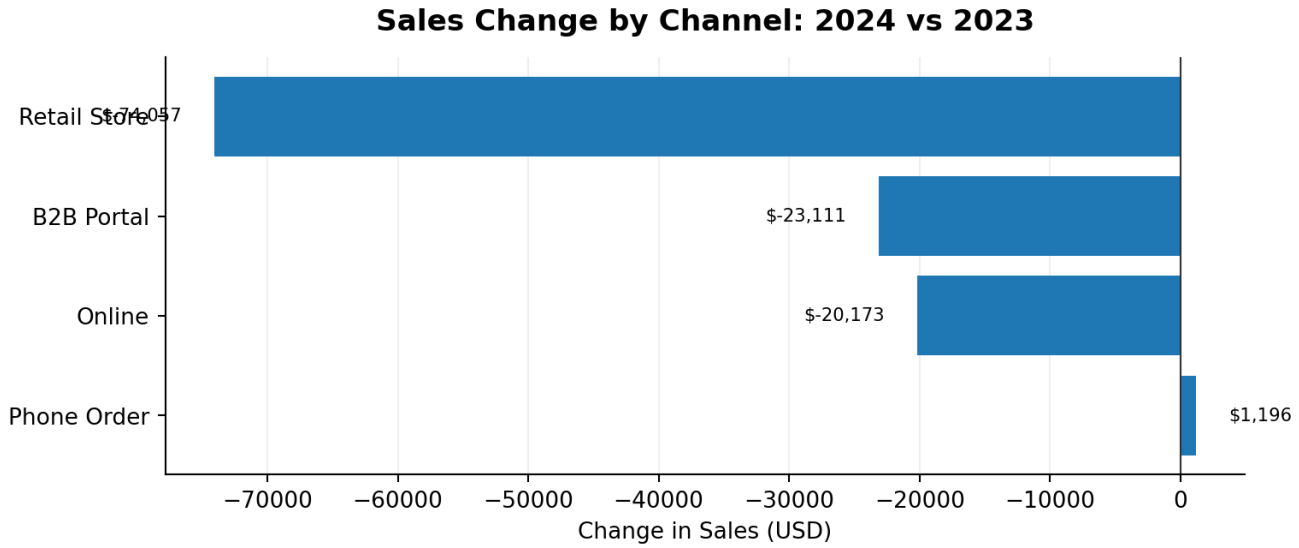
- December records the highest combined monthly sales.
- November and October also show strong performance.
- January and February have comparatively lower sales.
- Demand increases towards the end of the year.

Recommendation: Prepare inventory and marketing campaigns ahead of the high-demand period. Use historical monthly demand to plan stock levels and promotional budgets.

9. Root Cause Analysis: Why Did Sales Decline in 2024?

Sales declined from approximately $2.02M in 2023 to $1.91M in 2024. The decline was not limited to one category or region; multiple business areas contributed to the reduction.

Sales Channel Impact



- Retail Store experienced the largest decline, approximately $74.06K.
- B2B Portal declined by approximately $23.11K.
- Online sales declined by approximately $20.17K.
- Phone Orders recorded a small increase.

Other Contributing Factors

Product categories: Electronics declined by approximately $37.13K; Office Supplies by $31.18K; Furniture by $25.36K; and Appliances by $22.47K.

Geographical markets: Germany declined by approximately $50.41K, the UK by $41.71K and Canada by $40.92K. Australia declined by approximately $3.22K, while France increased by $4.35K and the United States by $15.76K.

Customer segments: Premium customers declined by approximately $30.40K, Returning Customer by $29.75K and Consumer by $24.62K.

Overall conclusion: The 2024 decline appears broad-based, with the largest identifiable channel-level reduction coming from Retail Stores. These results identify where sales declined, not necessarily why. Further investigation into customer behaviour, store performance, pricing, promotions and operational issues is required before assigning a definitive root cause.

Recommended actions

- Investigate underperforming retail stores.
- Analyse customer retention and repeat-purchase behaviour.
- Review product pricing and discount strategies.
- Investigate declining performance in Germany, the UK and Canada.
- Compare 2025 recovery patterns to identify successful strategies.

10. SWOT Analysis

STRENGTHS

- Strong electronics sales contribution
- Multiple sales channels
- Presence across several countries
- Profit margin of approximately 20%

WEAKNESSES

- Sales decline in 2024
- High dependence on Electronics
- Lower profitability at higher discounts
- Transaction return rate of 10.40%

OPPORTUNITIES

- Expand online sales
- Improve customer retention
- Strengthen underperforming markets
- Optimise inventory and delivery

THREATS

- Increasing discount pressure
- Product returns and delivery issues
- Revenue concentration in major categories
- Declining performance in certain markets

11. Business Growth & Scaling Strategy

1. Strengthen High-Performing Products

- Maintain stock availability for high-revenue products.
- Identify profitable product combinations.
- Introduce cross-selling and upselling strategies.

2. Improve Customer Retention

- Introduce loyalty programmes.
- Identify customers with declining purchase frequency.
- Provide targeted offers based on customer segments.

3. Optimise Discount Strategies

- Set discount limits based on profitability.
- Measure promotional effectiveness.
- Avoid excessive discounts on low-margin products.

4. Expand Sales Channels

- Improve online shopping experience.
- Strengthen B2B relationships.
- Investigate the reasons behind declining retail store sales.

5. Improve Operational Efficiency

- Reduce delivery delays.
- Improve product quality checks.
- Monitor return reasons.
- Use historical sales data for inventory planning.

12. Strategic Business Questions & Recommended Actions

Business Question	Recommended Action
Which products generate the most revenue?	Prioritise high-performing products and analyse their profitability.
Why did sales decline in 2024?	Investigate declining channels, categories and geographical markets.
Are discounts reducing profitability?	Introduce margin-based discount limits.
Which customers should be prioritised?	Focus on customer retention and high-value segments.
Which markets have growth potential?	Analyse regional demand and expand strategically.
How can returns be reduced?	Improve quality checks and delivery accuracy.
How can seasonal demand be managed?	Plan inventory and campaigns using historical monthly trends.
Which sales channels need improvement?	Focus on Retail Store performance and strengthen online sales.

13. Management Action Plan

Priority	Action	Suggested Target
High	Reduce transaction return rate	Below 8%
High	Improve discount profitability	Reduce loss-making discounts
High	Improve retail store performance	Recover 2023 sales levels
Medium	Improve customer retention	Increase repeat purchases
Medium	Improve delivery performance	Reduce Standard delivery time
Medium	Strengthen inventory planning	Reduce stock imbalance

These are proposed management targets, not measured outcomes.

14. Data Quality & Analytical Limitations

Several data quality checks and cleaning operations were performed before the business analysis.

Key data preparation considerations

- Duplicate and inconsistent records were reviewed during data preparation.
- Invalid quantity records were removed.
- Missing customer ratings and delivery-day values were retained where appropriate.
- Inconsistent text values were standardised.
- Numeric columns were checked and converted into suitable data types.
- Profit and sales calculations were checked for inconsistencies.

Analytical limitations

- Some records contain missing values in optional fields.
- Certain profit values do not reconcile exactly with sales minus cost.
- Repeated Order IDs mean distinct Order ID counts should not automatically be interpreted as true order counts.
- Return rate in this report is calculated at transaction level.
- The analysis identifies patterns and associations; it does not establish causation.
- The currency is not explicitly confirmed by the supplied column names; USD is used as an assumed reporting convention.

15. Conclusion

The Retail Sales Analysis project provides an overview of business performance across four years using SQL Server. The analysis highlights the importance of product performance, customer retention, pricing decisions, geographical markets, sales channels and operational efficiency.

Although sales declined in 2024, the recovery in 2025 indicates improved revenue performance. The business can build on this improvement by strengthening profitable product categories, improving customer retention, controlling discounts and reducing operational inefficiencies.

Overall, the project demonstrates how SQL can be used not only to retrieve and analyse data but also to identify business problems and support data-driven decision-making.

REPORTING NOTE

All figures and interpretations are carried forward from the supplied draft report. Before public release, reconcile the KPIs, category totals, currency convention and year-over-year comparisons against the final cleaned SQL table. The category chart is explicitly approximate.